Exercises 1

- Solve the Farmer's problem – to get $x_1=60$ and $x_2=20$
- Solve the problem we massaged

  Original problem is unbound

  Put an upper bound on the variables, e.g. 10

- Formulate and solve the equations to optimise sale

  A store has requested a manufacturer to produce pants and sports jackets.
  For materials, the manufacturer has 750 m² of cotton textile and 1,000 m² of polyester. Every pair of pants (1 unit) needs 1 m² of cotton and 2 m² of polyester. Every jacket needs 1.5 m² of cotton and 1 m² of polyester.
  The prices of a pair of pants and a jacket are fixed at 50 $ and 40$, respectively.

  What is the number of pants and jackets that the manufacturer must give to the stores so that these items obtain a maximum sale (incoming $)?

- Massage

$\max f = 3x_1 + 2x_2 - x_3$

$x_1 + 2x_2 + x_3 \leq 5$
$2x_1 + 4x_2 - x_3 \geq 1$
$x_1 \geq 0,\ x_2 \leq 0$